

Cisco Discovery Protocol (CDP)

Project Overview

This project was initiated to explore and document the functionality and operational aspects of the Cisco Discovery Protocol (CDP) within a Cisco-based network environment. The primary objective is to understand how CDP operates, how it facilitates device discovery, and how it contributes to network management and troubleshooting. The project focuses on the following goals:

- Examine how CDP enables Cisco devices to share and receive information at Layer 2.
- Understand the structure and content of CDP messages such as device ID, IP address, platform, and interface data.
- Analyze CDP's role in topology discovery, VLAN awareness, PoE management, and mismatch detection.
- Compare CDP versions (CDPv1 vs CDPv2) and identify the enhancements in version 2.
- Learn CDP configuration and monitoring commands including `cdp run`, `cdp enable`, `show cdp neighbors`, and `show cdp traffic`.
- Observe real-time CDP operation and message exchange between devices in a lab environment.

The project outcome provides a clear and detailed understanding of how CDP supports network visibility, device identification, and operational efficiency across Cisco devices in a Layer 2 topology.

Description

Cisco Discovery Protocol (CDP) is a proprietary Layer 2 network protocol developed by Cisco Systems. It allows Cisco devices to share information with other directly connected Cisco devices, enabling them to discover each other and exchange information. With CDP, the administrator can gather hardware and protocol information about neighboring devices, which can be helpful when troubleshooting or documenting the network.

Recommendations

To mitigate the risks associated with DHCP, the assessment team recommends the following steps:

- Enable CDP only when necessary
- Avoid enabling CDP on untrusted interfaces
- Be mindful of network traffic
- Ensure proper configuration
- Consider alternatives for multi-vendor environments
- Apply firmware/IOS updates for known CDP vulnerabilities (e.g., CVE-2020-3118).

References

<https://notes.networklessons.com/cdp-best-practices>
<https://www.networkacademy.io/ccna/network-services/cisco-discovery-protocol-cdp>
<https://learningnetwork.cisco.com/s/article/cisco-discovery-protocol-cdp-x>
[https://www.fs.com/es/glossary/cisco-discovery-protocol-\(cdp\)-68.html](https://www.fs.com/es/glossary/cisco-discovery-protocol-(cdp)-68.html)
[https://study-ccna.com/cisco-discovery-protocol-cdp-overview/#:~:text=CDP%20\(Cisco%20Discovery%20Protocol\)%20is,the%20locally%20attached%20Cisco%20equipment.](https://study-ccna.com/cisco-discovery-protocol-cdp-overview/#:~:text=CDP%20(Cisco%20Discovery%20Protocol)%20is,the%20locally%20attached%20Cisco%20equipment.)

Details

The protocol is enabled by default and requires no additional configuration to start working. It is plug-and-play. You power up a Cisco device, and it starts sending CDP messages out of every interface, as shown in the diagram below.

All Cisco devices transmit CDP packets periodically (default time interval value is 60 seconds though this is adjustable). These packets advertise a time-to-live (TTL) value in seconds, which indicates the number of seconds that the packet must be retained before it can be discarded (default value is 180 seconds).

- CDP packets are sent with a time-to-live value that is nonzero after an interface is enabled and
- With a time-to-live value of zero immediately before an interface is down. This provides quick state discovery.

All Cisco devices receive CDP packets, process them and cache the information in the packet. Cisco devices never forward a CDP packet. If any information changes from the last received packet, the new information is cached and the older information is discarded even if its time-to-live value has not yet expired.

CDP messages include the following information:

- Device name (hostname)
- Interface types and numbers (e.g., GigabitEthernet0/1)
- IP addresses of connected devices
- Cisco IOS version
- Capabilities (e.g., router, switch, etc.)
- Platform model (e.g., Cisco 9300 switch)
- VLAN information

At the data-link layer, the Cisco Discovery Protocol (CDP) plays several important roles in network operations, outlined below:

Topology Discovery: CDP allows devices, such as switches, routers, and IP phones, to identify other directly connected Cisco devices. This capability enables network engineers to acquire real-time information about the devices in the network and understand the physical layout.

Mismatch Detection: The protocol is useful for identifying issues like duplex mismatches and incorrect VLAN assignments, which can arise within the network.

VLAN Awareness: CDP helps IP phones determine the VOICE and DATA VLAN IDs configured on the switch port they're connected to, ensuring proper network configuration.

PoE Power Management: Through CDP, access points can communicate their power needs to the switch, which is crucial for receiving sufficient power, especially when multiple devices rely on Power over Ethernet (PoE) resources.

CDP versions:

CDPV1:- The initial which is capable only to collect device information connected to next end.

CDPV2: Is the most recent release of the protocol and provides more intelligent device features like instances of mismatch native VLAN IDs on 802.1Q trunks, and mismatch in duplex states between connecting devices.

CDP Frame Format:

CDP is assigned HDLC protocol type value 0x2000. A Cisco-proprietary SNAP value enumerates HDLC protocol type values so CDP can run on all media that support SNAP, such as LAN media, Frame Relay, and ATM.

The SNAP format is as follows:

- LLC—0xAAAA03
- Org ID—0x00000C
- HDLC protocol type—0x2000

CDP Configuration:

(config)#cdp run	Enables CDP globally on device
(config)# no cdp run	Disables CDO globally on device
(config)#cdp enable	Enables CDP on an interface device if CDP is enabled globally
(config)# no cdp enable	Disables CDP on an interface device.
(config)#cdp timer <seconds>	Specifies CDP packs transmission frequency. Default 60 sec.
(config)#cdp holdtime<seconds>	Specifies time limit for which a receiving device should hold information before discarding. Default 180 sec.

CDP Monitoring and Maintenance Commands:

- clear cdp counters
- clear cdp table
- show cdp
- show cdo entry device-name [protocol | version]
- show cdp neighbors [type number] [detail]
- show cdp traffic

We check the CDP table of a Cisco device using *show cdp neighbors:*

Example:

```
SW1# sh cdp neighbors
```

```
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge  
S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,  
D - Remote, C - CVTA, M - Two-port Mac Relay
```

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
SW2	Eth 0/1	134	R S I	Linux Uni	Eth 0/1
R1	Eth 0/0	145	R	Linux Uni	Eth 0/0

- The first column, "Device ID," shows the device's hostname.
- The second column, "Local Interface," shows switch 1's local interface that connects to the neighbor. For example, SW1's interface Eth0/1 connects to R1 while Eth0/0 connects to R1.
- The capability column shows if the device is a router, switch, or IP phone.
- The Platform identifies the specific model of the neighboring device. Since our devices are virtual IOL images, it shows Linux.
- The Port ID shows the neighboring device's local interface.

CDP Spoofing

In CDP spoofing, an attacker sends packet with multicast mac-address (01:00:0c:cc:cc:cc) as destination and various spoofed or fake mac-addresses as source. When a Cisco Device receives these frames it starts to add the information in CDP table and the table will start to build larger because the attacker may sends thousands of CDP frames to the device.

Affected Locations

- Switches
- Routers
- IP Phones and VoIP Devices
- Wireless Access Points
- Data Centers